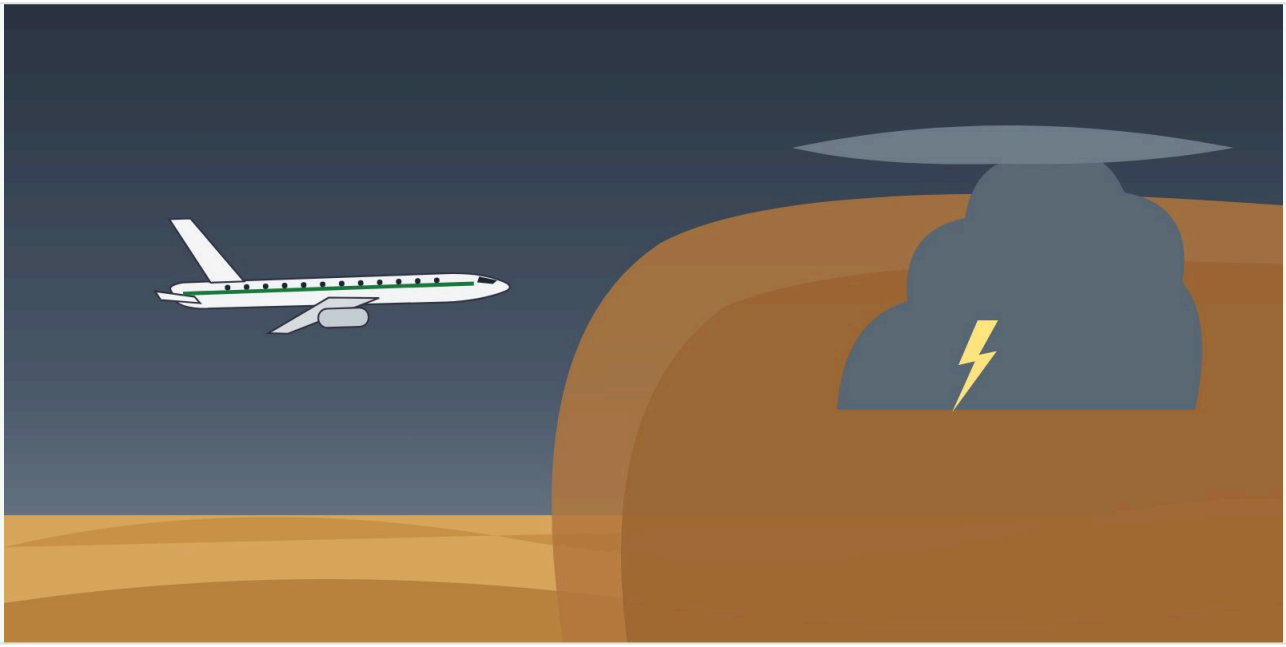
MODULE 2 · WEATHER, SAFETY AND PREPARATION

Lesson 04: Sand, Dust and Convective Weather

Lesson 4 of 6 · about 7 min

LEARNING OBJECTIVES

- Recognise the hazards of dust, sand and drifting sand.
- Apply precautions on the ground and in flight.
- Manage convective weather and gusty winds in summer.



Sand and dust can reduce visibility within minutes.

Dust and sand

- Watch the reports for `DU` (dust), `SA` (sand), `DRSA` (low drifting sand), `BLSA` (blowing sand), `SS` / `DS` (sandstorm / duststorm). Visibility can fall very quickly.
- During the walk-around in dusty conditions, check pitot probes, static ports, engine inlets, APU inlet and air conditioning inlets. Fit covers when the aircraft is parked.
- Minimise engine and APU running and high thrust on sand-covered surfaces to limit ingestion. Use the procedure for taxi and take-off on contaminated or sandy surfaces if published.
- Sand on the runway reduces braking — treat it like a contaminant and report braking action if worse than expected.

Convective weather and wind

- Summer thunderstorms (`CB`, `TCU`) bring gusts, windshear, turbulence and sudden dust storms. Avoid cells by a wide margin.
- **Gusty winds** — for example a reported wind of 18 kt gusting 28 kt — require a careful check of the company crosswind limits and the appropriate speed additives.
- Thermal turbulence close to the ground makes approaches less stable — be ready to go around.

- Dust devils and localised strong winds can affect take-off and landing.

Go-around in hot weather

Climb performance is lower in hot, high conditions. Brief the missed approach, the climb gradient and any obstacles. Know the go-around technique for your type, including engine-out, and practise it in the simulator in hot weather conditions.

KEY TAKEAWAYS

- DU, SA, DRSA, SS/DS: visibility can collapse — watch the trend.
- Protect inlets and probes; minimise ground running and high thrust on sand.
- Summer CB and gusts: wide avoidance, check crosswind limits and speed additives.
- Lower go-around performance in hot, high conditions — brief it.

KNOWLEDGE CHECK

What does DRSA in a METAR mean?

A. Dry runway, sand removed

B. Low drifting sand

C. Drizzle and sand

D. Dust storm

Answer: B. DRSA = low drifting sand.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.